

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“JNANA SANGAMA”, BELAGAVI - 590018**



**DEEP LEARNING MINI PROJECT REPORT (CS722T2C)
on
BRAIN TUMOR DETECTION ON MRI IMAGES USING CNN
*Submitted by***

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In partial fulfillment of the requirements for the VII semester
**BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE & ENGINEERING**

Under the Guidance of
Ms. Shiji Abraham
Assistant Professor
Department of CSE



**SAHYADRI
College of Engineering & Management
An Autonomous Institution
MANGALURU
2026-27**



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EVALUATION RUBRICS

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Sl. No.	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
	Total					



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DECLARATION

We hereby declare that the entire work embodied in this Mini Project Report titled “**Brain Tumor Detection on MRI Images Using CNN**” has been carried out by us at Sahyadri College of Engineering & Management, Mangaluru under the supervision of **Ms. Shiji Abraham**, in partial fulfillment of the requirements for the VII semester of **Bachelor of Engineering in Computer Science & Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

Magnetic resonance imaging is central to the assessment of intracranial tumours because it provides complementary soft-tissue contrast without ionising radiation. Interpretation remains difficult: tumour boundaries may be diffuse, enhancement is heterogeneous, oedema can extend beyond the visible core, and a single study contains many slices across several sequences. This mini project develops SYSTEM.ONCO, a reproducible research prototype for automated glioma localisation and volumetric segmentation using a three-dimensional convolutional neural network. The system accepts four co-registered NIfTI volumes—T1, contrast-enhanced T1, T2 and FLAIR—validates their geometry, normalises non-zero intensities and applies a pretrained MONAI SegResNet by Gaussian sliding-window inference. The output channels represent Whole Tumor, Tumor Core and Enhancing Tumor. A deterministic label map, whole-tumour probability volume and regional three-dimensional Grad-CAM explanation are written back in the original spatial frame.

A FastAPI service exposes the local study catalogue, upload validation, bounded analysis jobs, preparation protocol and downloadable outputs. A Next.js interface uses NiiVue/WebGL2 to display axial, coronal, sagittal, multiplanar and rotatable three-dimensional views. Reviewers can scan the volume slice by slice, apply a cutaway plane, change sequences, adjust overlay opacity, compare the CNN output with the reference mask and move the crosshair to reported tumour centroids. Pipeline and data-preparation details are placed in separate tabs so the main dashboard remains focused on evidence needed for review.

The implemented system was exercised on three local BraTS 2023 adult-glioma studies. Whole-Tumor Dice coefficients were 0.918, 0.458 and 0.878, with a mean of 0.751. The low-performing case is retained because it demonstrates why mean sigmoid probability must not be presented as diagnostic accuracy. Kaggle and Mendeley image collections are documented as sources for a separate two-dimensional classification experiment, but they are not mixed with the four-sequence volumetric inference pipeline. The prototype is intended for education, reproducible experimentation and visual inspection; it is not a medical device and does not determine tumour type, grade, prognosis or treatment.

Keywords: brain tumour, MRI, 3D CNN, SegResNet, BraTS, NIfTI, Grad-CAM, NiiVue, FastAPI.

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1. Introduction

1.1 Background

Brain tumours are structurally and biologically heterogeneous lesions. Their appearance varies with tumour lineage, grade, treatment history, acquisition protocol and the physical response of surrounding tissue. MRI is preferred for soft-tissue assessment because different pulse sequences reveal different aspects of anatomy and pathology. A radiologist therefore reasons across a registered set of volumes rather than from a single image. This project adopts the same multimodal principle: the CNN receives T1, T1c, T2 and FLAIR volumes as four aligned channels and predicts spatially corresponding tumour regions.

The term detection is used in the project title in the practical sense of finding and localising abnormal tumour tissue. The implemented task is technically semantic segmentation: every voxel is assigned to background or a tumour-related region. This distinction matters. A two-dimensional classifier answers whether a selected image belongs to a class, whereas a volumetric segmenter produces a three-dimensional region that can be measured, overlaid on the source MRI and reviewed at every slice.

1.2 MRI and Tumour Sub-regions

T1-weighted imaging provides anatomical structure; post-contrast T1 highlights regions where contrast uptake suggests blood–brain barrier disruption; T2 is sensitive to fluid-rich tissue; and FLAIR suppresses free-fluid signal so oedema and infiltrative abnormality remain conspicuous. BraTS conventions organise labels into clinically useful composite regions. Whole Tumor includes all labelled tumour-related tissue, Tumor Core combines enhancing and non-enhancing core, and Enhancing Tumor isolates the enhancing component. These regions are nested and their volumes must not be added together.

SYSTEM.ONCO preserves this hierarchy throughout inference, measurement and presentation. It reports voxel-derived volumes, centres of mass, overlap scores where a reference exists, and an attribution map that indicates regions influencing the chosen model response.

1.3 Motivation and Contribution

The main contribution is an end-to-end, inspectable workflow rather than a standalone notebook. Data provenance, preparation, model revision, geometry preservation, job status, output assets and interface behaviour are all explicit. The dashboard keeps the MRI and measured findings together, while separate Data & Preparation and Pipeline tabs expose the behind-the-scenes protocol. This separation supports clarity without hiding technical limitations.

2. Literature Survey

2.1 CNN Foundations and Volumetric Models

U-Net established an encoder–decoder structure in which down-sampling layers learn increasingly abstract features and skip connections return high-resolution spatial information to the decoder [5]. The original network was designed for two-dimensional biomedical images, but its structural idea became the basis of many medical segmentation systems. Çiçek et al. extended the formulation to three dimensions, allowing convolutional filters to operate jointly across width, height and depth [6]. This change is important for brain MRI because the appearance of a lesion in one slice is constrained by neighbouring anatomy.

V-Net introduced an end-to-end volumetric network and optimised an overlap-based objective suited to severe foreground–background imbalance [7]. Pereira et al. demonstrated that intensity normalisation and small convolutional kernels could support MRI tumour segmentation [8]. DeepMedic combined multiple spatial scales so the model could preserve local boundary detail while observing a larger anatomical context [9]. Together, these works established the major design choices still visible in modern pipelines: multimodal inputs, 3-D context, multi-scale encoding, overlap-aware evaluation and careful preprocessing.

Study	Main contribution	Relevance to this project
U-Net [5]	Encoder–decoder with skip connections	Provides the basic localisation architecture.
3D U-Net [6]	Volumetric convolutions and sparse-label learning	Motivates processing the complete NIfTI volume.
V-Net [7]	End-to-end 3-D segmentation and Dice objective	Addresses tumour/background class imbalance.
Pereira et al. [8]	CNN segmentation with MRI intensity normalisation	Supports per-channel non-zero z-score normalisation.
DeepMedic [9]	Dual-resolution 3-D context	Shows the value of local and global evidence.

These methods are not directly interchangeable: reported performance depends on cohort, label definition, data split and post-processing. Accordingly, the present work treats published results as design evidence rather than as a score that can be transferred to the three local cases.

2.2 SegResNet, Self-configuration and Explainability

Myronenko proposed an autoencoder-regularised three-dimensional encoder–decoder that won the BraTS 2018 challenge [10]. The segmentation path uses residual blocks, while the auxiliary reconstruction branch regularises training when labelled data are limited. The released MONAI bundle used in this project provides the trained SegResNet inference network with four input channels, three sigmoid outputs, 16 initial filters, encoder block counts of 1–2–2–4 and decoder block counts of 1–1–1 [18]. Pinning the exact model revision improves reproducibility and prevents silent changes in weights or configuration.

nnU-Net demonstrated that systematic, data-dependent configuration can outperform many hand-designed architectures [11]. Its significance is methodological: preprocessing, patch size, resampling and validation strategy can matter as much as architectural novelty. Grad-CAM uses gradients flowing into a convolutional feature layer to form a coarse localisation map [12]. In this project the method is adapted to a 96-voxel three-dimensional crop and a tumour-core score. The map is therefore an attribution for one regional model response; it is not a probability map, boundary truth or independent diagnosis.

2.3 Evidence Synthesis and Research Gap

A systematic review and meta-analysis by Kouli et al. found that deep learning outperformed traditional machine learning for several detection and sub-compartment tasks, but only a minority of included studies reported external validation [23]. Reviews also emphasise heterogeneous preprocessing, limited multi-centre validation and inconsistent reporting as barriers to clinical generalisation [24], [25]. This evidence supports a cautious interpretation of small demonstration studies.

The practical gap addressed here is the distance between a model output and an auditable user workflow. Many student projects stop at a classification score or a static prediction image. SYSTEM.ONCO instead retains geometry, exposes the exact preparation protocol, distinguishes probability from accuracy, shows multiple MRI sequences, permits layer-by-layer and three-dimensional review, and keeps reference comparison outside inference.

Literature concern	Design response in SYSTEM.ONCO
Loss of 3-D context	Four full volumes and 3-D sliding-window inference.
Opaque preprocessing	Protocol table and pinned channel order/model revision.
Uncalibrated confidence	Probability is labelled as model output, not diagnostic accuracy.
Static result images	NiiVue slice scanning, multiplanar views and 3-D cutaway.
Weak traceability	Persistent jobs, downloadable NIfTI assets and source metadata.

3. Problem Formulation

3.1 Problem Description

A typical BraTS study contains four $240 \times 240 \times 155$ volumes. Manual review therefore involves hundreds of two-dimensional planes and cross-comparison between modalities. Tumour margins may be indistinct, small enhancing regions may occupy a tiny fraction of the field of view, and acquisition intensity is not standardised across subjects. An automated system must handle these properties while preserving enough context for a reviewer to inspect mistakes.

The engineering problem is broader than running a neural network. Input volumes must belong to the same subject, have identical shape and affine geometry, contain finite non-empty data and use a valid spatial unit. Model input channels must follow the checkpoint contract. Large volumes must be processed within local hardware limits. Outputs must be mapped to the original grid so overlays do not drift. Finally, the interface must communicate results without overstating what the model can establish.

3.2 Problem Statement

Design and implement a CNN-based research application that accepts aligned multimodal brain MRI volumes, segments glioma-related regions, produces a regional explanation, extracts reproducible spatial measurements and presents the complete result in an interactive browser-based three-dimensional viewer.

3.3 Objectives

- Validate T1, T1c, T2 and FLAIR NIfTI inputs before inference.
- Run a pinned pretrained 3-D SegResNet with CUDA, Apple MPS or CPU fallback.
- Preserve original dimensions, affine matrix, qform, sform and datatype intent in outputs.
- Generate segmentation, whole-tumour probability and regional Grad-CAM volumes.
- Calculate tumour volumes, centroids and reference Dice scores when labels are available.
- Support continuous slice scanning, multiplanar inspection, rotatable 3-D rendering and oblique cutaway controls.
- Separate dashboard evidence from data provenance and pipeline implementation details.
- Document limitations so the prototype is not interpreted as a clinical diagnostic system.

3.4 Research Questions

The project asks whether a pretrained volumetric CNN can be integrated into a transparent local workflow; whether NIfTI geometry can be preserved across preprocessing and output generation; whether multiple result types can be distinguished clearly; and whether interactive review reveals failure cases that a single summary score would conceal.

3.5 Scope, Assumptions and Requirements

The implemented scope is adult-glioma sub-region segmentation from four registered MRI sequences. It excludes DICOM ingestion, registration of misaligned scans, tumour grading, molecular prediction, treatment recommendation, authentication and hospital deployment. BraTS reference labels are used only after inference for evaluation. Kaggle and Mendeley image datasets remain a documented two-dimensional classification track and are not treated as interchangeable with volumetric BraTS data.

ID	Functional requirement	Acceptance evidence
F1	List prepared studies and their provenance	Data tab shows source, dimensions and label availability.
F2	Upload four NIfTI sequences	Server validates suffix, size, shape, affine and contents.
F3	Run CNN analysis asynchronously	Queued job exposes stage, progress and terminal status.
F4	Display all MRI orientations	Axial, coronal, sagittal, multiplanar and 3-D controls work.
F5	Scan the brain level by level	Slice interaction and 3-D cutaway depth/angle controls are available.
F6	Switch overlays	Segmentation, probability, Grad-CAM and reference are selectable.
F7	Report quantitative results	Volumes, centroids, mean probability and Dice are returned.
F8	Download aligned outputs	Mask, probability and Grad-CAM NIfTI files are served.
Quality attribute	Requirement	
Correctness	Reject incompatible inputs; preserve geometry; keep reference labels outside inference.	
Reproducibility	Pin dataset and model revisions; record device, shape and model name.	
Usability	Visible status, labelled controls, keyboard focus and high-contrast states.	
Performance	Use 96^3 sliding windows, batch size one and bounded background execution.	
Privacy	Run on loopback; uploaded studies remain on the local computer.	
Safety	Avoid diagnostic claims and retain explicit clinician-review messaging.	

4. Project Design and Implementation

4.1 End-to-End Architecture

SYSTEM.ONCO — END-TO-END PROCESS ARCHITECTURE

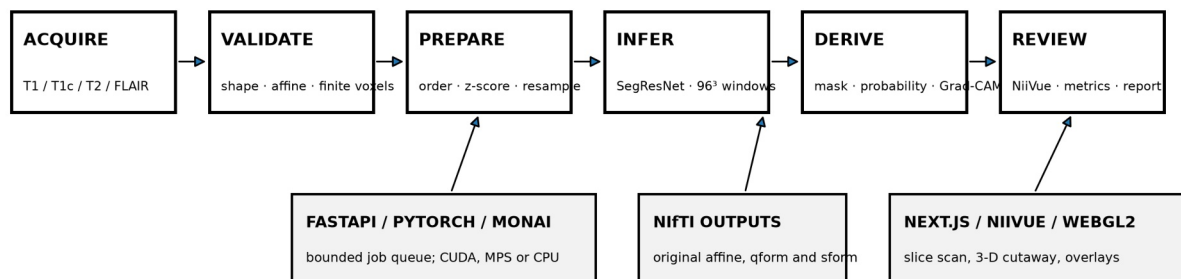


Figure 1. End-to-end SYSTEM.ONCO architecture. Source: Authors’ implementation, based on MONAI SegResNet, NIfTI and NiiVue [10], [13], [15], [16], [18].

The architecture has three explicit boundaries. The data boundary accepts only a complete four-sequence study and produces a validated internal record. The inference boundary converts those volumes into the checkpoint channel order, runs the CNN and derives spatial outputs. The presentation boundary serves immutable result assets to the browser and visualises them against the source anatomy.

FastAPI owns long-running work and file access; Next.js owns interaction and report presentation; NiiVue owns WebGL rendering. This separation keeps GPU or CPU inference outside the React lifecycle and prevents browser layout changes from altering scientific data. Each completed job records the model revision, execution device, input shape, affine matrix, label definitions, heatmap scope and asset URLs.

Layer	Primary responsibility	Failure handled
Data	Catalogue, upload, validation and provenance	Missing sequence, invalid affine, empty or non-finite volume.
Inference	Normalisation, SegResNet, post-processing and Grad-CAM	Unavailable weights, unsupported MPS operation, job failure.
Service	Queue, persistence, progress and NIfTI delivery	Queue saturation, restart interruption, missing output.
Interface	3-D review, measurements, reports and downloads	Offline service, loading state and absent result.

4.2 Dataset Description and MRI Modalities

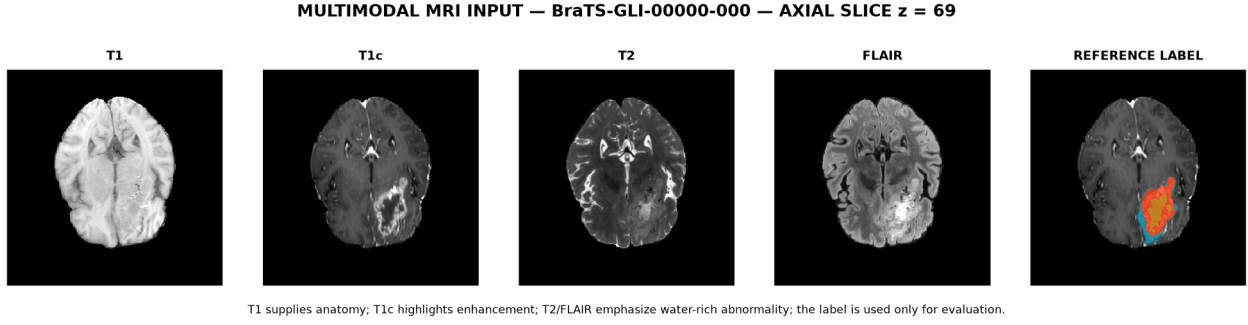


Figure 2. Four co-registered MRI sequences and reference annotation for case BraTS-GLI-00000-000. Source: BraTS 2023 GLI mirror [17]; visualisation by the authors.

Three BraTS 2023 GLI studies were downloaded from the pinned MedOtter Hugging Face revision. Each study contains T1, T1c, T2 and FLAIR NIfTI volumes plus a dense reference annotation. These cases are suitable for an end-to-end demonstration because the modalities are already aligned and skull-stripped. They are not an independent benchmark: the deployed model was trained on BraTS 2018, and possible subject overlap between challenge editions has not been excluded.

The Kaggle collection supplied by Nickparvar contains 7,200 two-dimensional images across glioma, meningioma, pituitary and no-tumour classes [19]. The Mendeley source contains 12,064 contrast-enhanced T1 images in the same broad classification setting [20]. Both are useful for a separate classification experiment, but neither provides the aligned four-volume NIfTI structure and voxel-wise labels required by the present 3-D segmentation pipeline.

Resource	Format	Role in this project
BraTS 2023 GLI [17]	Four 3-D NIfTI sequences + mask	Inference demonstration and overlap evaluation.
MONAI bundle [18]	SegResNet weights + configuration	Pinned pretrained segmentation baseline.
Kaggle MRI [19]	7,200 labelled 2-D images	Documented source for future classification only.
Mendeley MRI [20]	12,064 labelled 2-D CE-T1 images	Documented source for future classification only.

4.3 Data Preparation Protocol

Preparation is deterministic and recorded by the service. The model never receives the reference segmentation. Validation occurs before the study is admitted to inference, preventing visually plausible but spatially invalid overlays.

Step	Operation	Implementation detail
01	Acquire	Collect T1, T1c, T2 and FLAIR from one subject; retain source provenance.
02	Validate dimensions	Require a 3-D shape and no more than 32 million voxels per volume.
03	Validate geometry	Require equal shapes and affine matrices within 10^{-4} tolerance.
04	Validate values	Reject non-finite, empty or unsupported spatial-unit data.
05	Reorder channels	Convert catalogue order T1–T1c–T2–FLAIR to checkpoint order T1c–T1–T2–FLAIR.
06	Normalise	Apply independent z-score normalisation to non-zero voxels in each channel.
07	Resample if required	Convert non-1 mm inputs to 1 mm isotropic working space; invert to the original grid after inference.
08	Infer	Use 96^3 Gaussian windows with 25% overlap and batch size one.
09	Discretise	Apply sigmoid and threshold 0.5; enforce ET, then TC, then WT label precedence.
10	Restore and save	Write uint8 mask and float32 maps with original affine, qform and sform.

4.3.1 Leakage and Provenance Controls

Reference masks are loaded only after a prediction file exists. This prevents label leakage into preprocessing, inference or Grad-CAM. Dataset revision b032d353a3e80911a5f850bc54e6fb575298a354 and model revision 370f7f9d062745fbac445e7fe6d6616d35df04ec are recorded so another researcher can recover the same assets. A future controlled study should additionally use patient-level train, validation and test partitions and screen for overlap across BraTS editions.

4.4 CNN Model and Inference Procedure

SegResNet is a residual three-dimensional encoder–decoder. The encoder progressively reduces spatial resolution while increasing the number of feature channels. Residual connections allow a block to learn a correction to its input and reduce optimisation difficulty in deeper networks. The decoder restores spatial resolution and combines semantic context with localisation information. The configured network begins with 16 filters, accepts four channels and emits three overlapping logits corresponding to Tumor Core, Whole Tumor and Enhancing Tumor [10], [18].

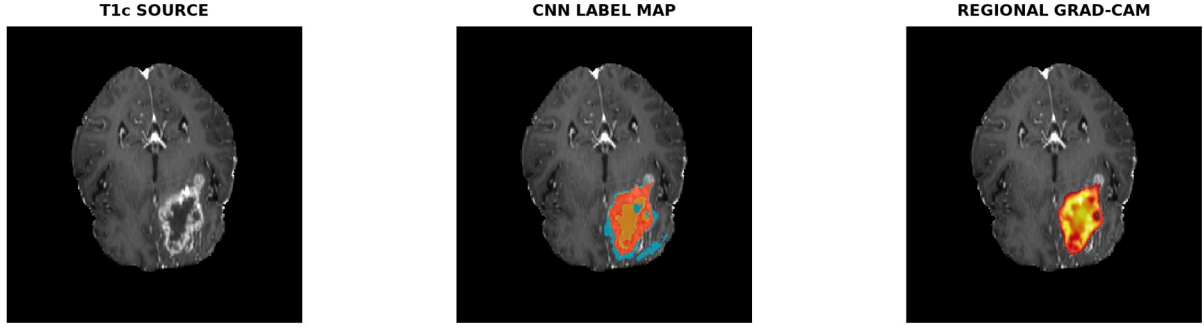
The full $240 \times 240 \times 155$ volume is larger than the 96^3 inference region. Gaussian sliding-window inference tiles the volume with 25% overlap. Predictions near a patch centre receive more weight than predictions near its edge, which reduces visible seams when tiles are combined. Batch size one bounds memory use. CUDA is selected when available, otherwise Apple MPS is attempted, and CPU is the final fallback. If an unsupported Apple GPU operation raises an error, inference is retried on CPU rather than returning a partial result.

The three output channels are independent sigmoid probabilities rather than a softmax partition. Thresholding at 0.5 converts them to binary regions. Because the regions overlap, a deterministic label map assigns Enhancing Tumor first, then non-enhancing Tumor Core, then oedema/Whole Tumor. The saved probability volume contains the Whole-Tumor channel and remains continuous for opacity-based review.

Parameter	Configured value	Reason
Input	$1 \times 4 \times D \times H \times W$	Four multimodal volumes for one subject.
Patch	$96 \times 96 \times 96$ voxels	Matches the bundle while limiting memory.
Overlap	25%	Blends neighbouring predictions.
Outputs	TC, WT, ET logits	Matches BraTS composite regions.
Activation	Sigmoid	Supports overlapping regions.
Threshold	0.5	Bundle-compatible deterministic discretisation.
Device	CUDA $\rightarrow$ MPS $\rightarrow$ CPU	Uses available acceleration with safe fallback.

4.5 Explainability and Quantitative Metrics

PREDICTION AND EXPLANATION ARE DISTINCT OUTPUTS



The label map is a thresholded segmentation; Grad-CAM is a regional attribution map and is not a second segmentation.

Figure 3. Source MRI, thresholded CNN segmentation and regional 3-D Grad-CAM. Source: BraTS 2023 GLI [17] and authors’ implementation based on Grad-CAM [12].

The explanation module selects a 96^3 crop centred on the predicted Tumor Core, attaches MONAI Grad-CAM to the final spatial decoder layer and differentiates the mean tumour-core logit. Positive values are normalised to $[0,1]$, values below 0.15 are suppressed, and the crop is inserted into a zero-valued full-size volume. This design keeps the explanation computationally bounded, but it also limits interpretation: absence of colour outside the crop means “not evaluated by this attribution operation,” not “irrelevant to the network.”

Tumour volume is computed as the number of voxels in a region multiplied by $|\det(A_{3 \times 3})|$, the absolute determinant of the affine spatial block. Results are converted from mm^3 to mL by division by 1,000. A region centroid is the mean index of all foreground voxels and is also transformed into millimetre coordinates using the affine. Dice overlap is $2|P \cap G|/(|P| + |G|)$, where P is the prediction and G is the reference. Dice evaluates overlap but does not describe boundary distance, lesion count or clinical acceptability [22].

Output	Meaning	Explicit limitation
Mean WT probability	Average sigmoid output inside predicted WT	Not calibrated diagnostic confidence.
Dice coefficient	Prediction/reference overlap	Requires a label; insensitive to some boundary errors.
Grad-CAM	Features influencing regional TC response	Coarse attribution, not a segmentation.
Volume	Voxel count $\times$ physical voxel volume	Depends on segmentation correctness.

4.6 FastAPI Backend and Job Lifecycle

The backend is a local FastAPI application with a single-worker ThreadPoolExecutor. A bounded queue avoids concurrent model runs exhausting memory. Each job moves through queued, running, completed or failed states. Progress callbacks persist the current stage and percentage to a temporary JSON file that is atomically renamed, reducing the chance of a partially written status record.

Uploads are streamed in 1 MB chunks and limited to 128 MB per sequence. Filenames do not determine study identity; the service creates a random identifier and validates the saved NIfTI files before adding a catalogue record. CORS is restricted to localhost origins. Completed assets are served only when the job record confirms completion and the requested asset name belongs to a fixed allow-list.

A server restart converts unfinished persisted jobs into an interrupted failure rather than falsely showing them as active. The API therefore favours explicit failure over silent ambiguity. The optional language-model report route is server-side and receives derived metrics rather than raw MRI voxels.

Method and route	Purpose	Response
GET /health	Model/device readiness	Availability, device, model and revision.
GET /studies	Local catalogue	Study ID, source, shape, spacing and label flag.
POST /studies/upload	Import four sequences	Validated study or HTTP 422 error.
POST /studies/{id}/analyze	Queue inference	HTTP 202 job record.
GET /jobs	Monitor work	Newest-first persisted job list.
GET /jobs/{id}/assets/{name}	Download result	Aligned NIfTI segmentation, probability or Grad-CAM.
GET /protocol	Expose method	Source revision and eight preparation stages.

4.6.1 Error Handling

Input errors produce actionable messages for missing modalities, invalid shapes, misaligned affines, unsupported units, empty volumes and excessive file size. Infrastructure errors identify missing model weights, queue saturation or interrupted work. These messages are surfaced in the dashboard with a retry action.

4.7 Interactive NiiVue Viewer

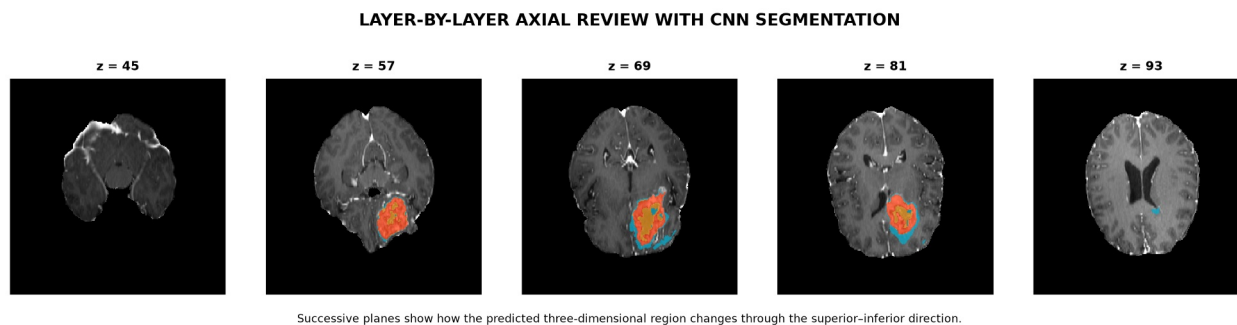


Figure 4. Layer-by-layer axial scan with the CNN segmentation overlay. Source: BraTS 2023 GLI [17] and authors' prototype.

NiiVue parses NIfTI data in the browser and renders it as WebGL2 textures [16]. The component is loaded only on the client to avoid server-side WebGL errors. A stable canvas owns one NiiVue instance; React controls pass view mode, sequence, overlay and opacity changes through the component boundary without recreating the entire page.

Five view selectors provide axial, coronal, sagittal, multiplanar and volume-rendered modes. In slice modes the reviewer can move through every level by dragging or scrolling. In 3-D mode the brain can be orbited and zoomed; an oblique cutaway plane exposes internal layers. A depth slider and play control support continuous scanning when cutting is enabled. Crosshair changes update the X, Y and Z voxel indices and current intensity in the compact HUD.

Overlay choices remain semantically distinct: the CNN label map uses categorical colours, Whole-Tumor probability is continuous, Grad-CAM uses a hot attribution map, and the dataset reference is available only for labelled cases. Opacity is adjustable so anatomy remains visible beneath the overlay. Interactive report terms set a crosshair target, creating a traceable bridge from a written finding to the corresponding voxel.

Interaction	Review purpose
Sequence switch	Compare anatomy, enhancement and fluid-sensitive abnormality.
Slice scan	Inspect entry, maximum extent and exit of a region across levels.
Multiplanar view	Confirm one coordinate in three orthogonal planes.
3-D cutaway	Explore internal boundaries without hiding the surface context.
Overlay/opacity	Compare model evidence with source MRI and reference.

5. Prototype and Results

5.1 Integrated Dashboard

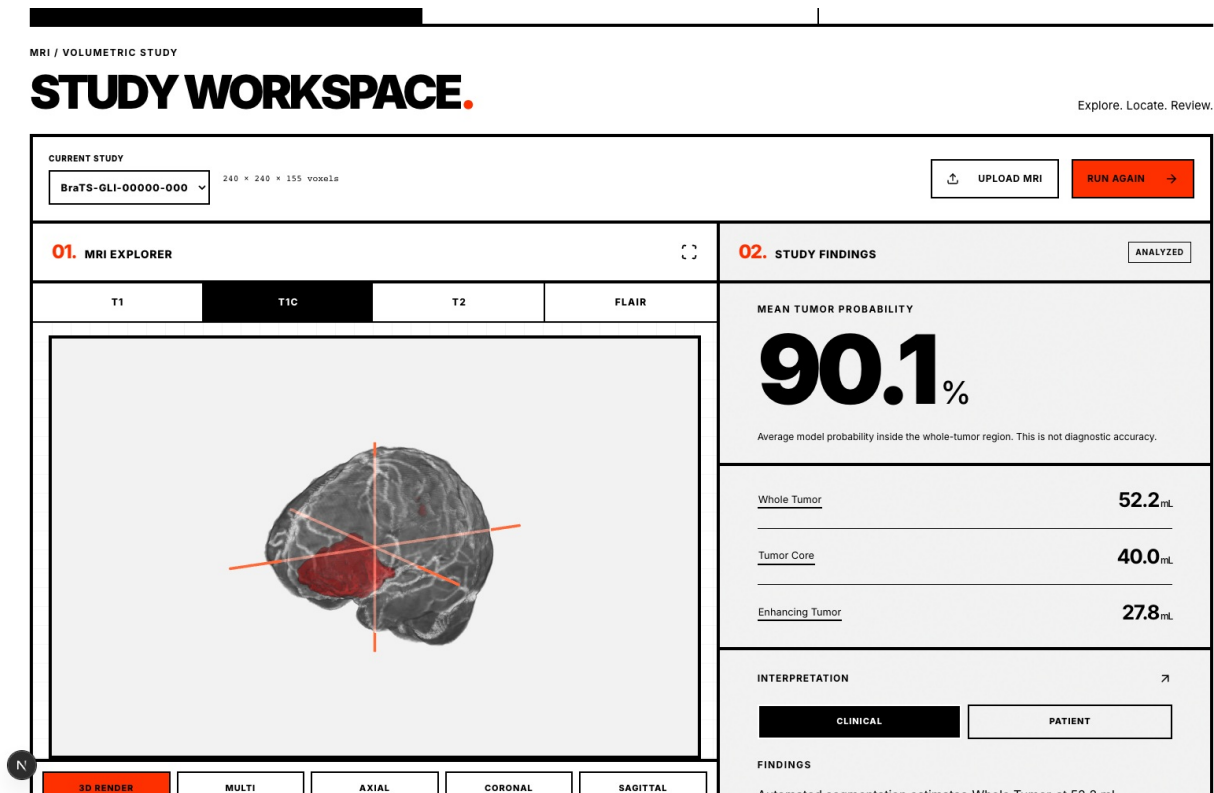


Figure 5. SYSTEM.ONCO dashboard showing the 3-D MRI, CNN overlay and measured findings for case BraTS-GLI-00000-000. Source: Authors' prototype.

The dashboard uses a 7:5 asymmetric grid. The larger observation column prioritises direct image review; the findings column reports model state, mean Whole-Tumor probability, nested region volumes and interpretation limits. The monochrome Swiss layout makes structure visible through borders and spacing, while red is reserved for active state and tumour-related emphasis.

The toolbar binds the selected catalogue study to both columns. “Analyze study” creates a backend job; progress is shown by stage and percentage; completed assets immediately become available to the viewer and download links. The page avoids displaying preparation details beside the MRI, because those details are important for audit but distract from the primary review task.

5.2 MRI Explorer and Interaction Controls

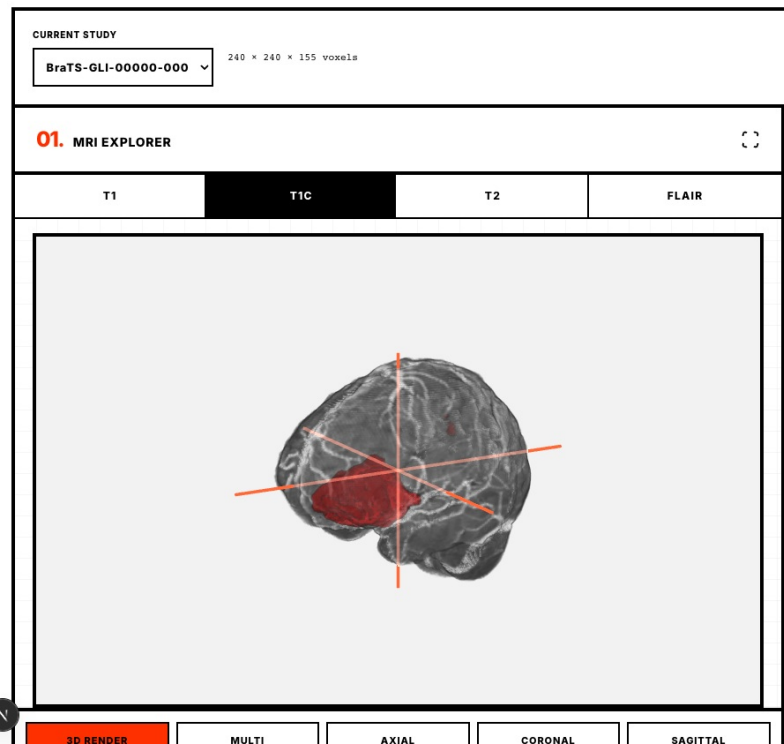


Figure 6. Prototype MRI explorer with sequence selectors, 3-D rendering and spatial crosshair. Source: Authors' prototype.

The explorer begins with T1c because contrast enhancement is visually informative for the selected example, but T1, T2 and FLAIR remain one click away. The current implementation loads one base volume and one selected overlay at a time, which limits GPU memory and keeps interactions smooth. The viewer preserves volume orientation from the NIFTI header rather than assuming that array indices correspond directly to anatomical directions.

The 3-D rendering exposes tumour position relative to the brain surface. Orthogonal crosshair lines locate the current voxel, and the same coordinate is available to the text-to-viewer bridge. For systematic inspection, the reviewer switches to an orthogonal plane and advances one level at a time; for spatial context, the reviewer returns to 3-D and uses the cutaway depth control. These two modes are complementary rather than competing visualisations.

Control group	States	Result
Sequence	T1 / T1c / T2 / FLAIR	Replaces the anatomical base volume.
View	3-D / Multi / Axial / Coronal / Sagittal	Changes projection while retaining the study.
Cutaway	Enable, depth, angle, play	Scans internal 3-D layers.
Overlay	Mask / probability / Grad-CAM / reference	Selects one evidence layer.
Opacity	0–100%	Balances anatomy and overlay visibility.
Crosshair	On / off / target	Links coordinates, intensity and report terms.

5.3 Findings, Data and Pipeline Views

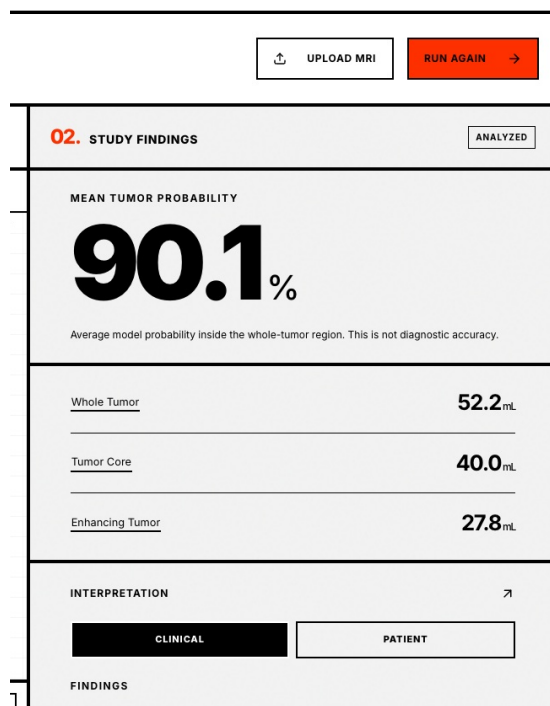


Figure 7. Findings panel with probability label, tumour volumes and report-audience control. Source: Authors’ prototype.

The findings panel deliberately states “mean tumour probability” rather than “accuracy.” Region names are interactive targets: selecting or hovering a term can move the viewer crosshair to that region’s centroid. Clinical text uses technical language and explicit limits; patient text is intended to use simpler language without adding unsupported conclusions. When the Gemini API key is absent or generation fails, a measured local summary remains available.

Two additional tabs separate audit information from the review dashboard. Data & Preparation lists local studies, external dataset references, model provenance and the first preparation stages. Pipeline lists job history, output availability and the complete protocol from acquisition through reporting. This structure answers the user’s requirement that behind-the-scenes processing be available without crowding the principal visual workspace.

Tab	Information shown	Reason for separation
Dashboard	MRI, overlays, metrics and interpretation	Supports focused case review.
Data & Preparation	Studies, datasets, model and preparation protocol	Supports provenance and reproducibility.
Pipeline	Jobs, stages, Dice and downloadable masks	Supports operational audit and troubleshooting.

5.3.1 Accessibility and Responsiveness

Controls have labels and visible focus states, interactive targets meet minimum sizes, status changes use text as well as colour, and motion can be reduced. On smaller screens the asymmetric grid stacks vertically while borders, contrast and full-width controls remain. The WebGL canvas retains a bounded height so report content remains reachable.

5.4 Experimental Evaluation

Inference was executed locally on Apple Silicon through PyTorch MPS. All source and output volumes had shape $240 \times 240 \times 155$ and 1 mm^3 voxels. The same pretrained weights, threshold and preparation protocol were used for all cases. Reference masks were introduced only after prediction to compute Dice.

Case	WT mL	TC mL	ET mL	WT Dice	TC Dice	ET Dice	Mean WT prob.
00000	52.2	40.0	27.8	0.918	0.937	0.894	90.1%
00002	57.0	10.8	6.5	0.458	0.473	0.429	83.0%
00003	78.0	38.9	19.6	0.878	0.965	0.880	86.0%
Mean	62.4	29.9	18.0	0.751	0.792	0.734	86.4%

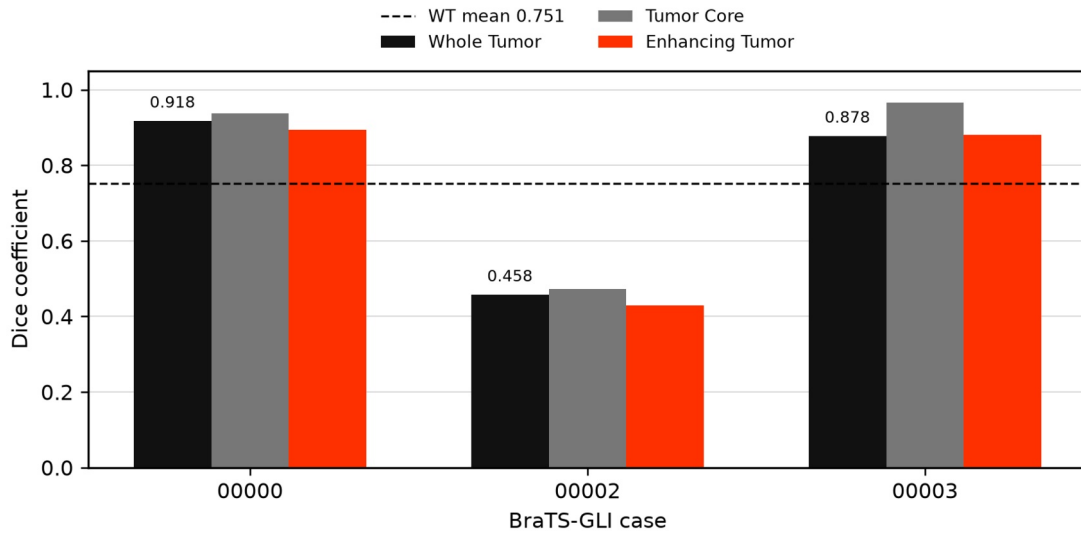


Figure 8. Dice coefficients for Whole Tumor, Tumor Core and Enhancing Tumor across three demonstration cases. Source: Authors' experiments.

Cases 00000 and 00003 achieved strong overlap for all three reported regions. Case 00002 was substantially weaker, with Whole-Tumor Dice 0.458 despite mean Whole-Tumor probability 83.0%. This disagreement is the central evaluation finding: probability magnitude inside the predicted region does not establish spatial correctness. A reviewer who saw only a high percentage could be misled, whereas the overlay and reference comparison reveal the failure immediately.

All generated assets retained the input affine matrix and dimensions, allowing direct overlay in NiiVue. The functional path from catalogue selection through analysis, progress, visual review and download completed for all three studies.

6. Discussion, Conclusion and Future Scope

6.1 Discussion

The experiment demonstrates technical integration rather than population-level clinical performance. A mean Whole-Tumor Dice of 0.751 across three cases is informative only as a local software check. The range from 0.458 to 0.918 confirms that performance can vary sharply between subjects. This aligns with review evidence that external validation, acquisition diversity and transparent preprocessing remain major weaknesses in published brain-tumour AI research [23]–[25].

Compared with a simple image classifier, the system provides a more useful spatial result: a reviewer can inspect boundaries, calculate physical volume and move through every plane. Compared with a model-only segmentation notebook, it adds input validation, persistent job state, geometry-preserving outputs, model provenance, explanation, reference comparison and a usable viewer. The cost is greater implementation complexity and the need to define each output carefully.

6.2 Limitations

- Three BraTS cases are insufficient for statistical conclusions and may overlap with the model’s broader training lineage.
- SegResNet is a glioma segmentation model; it is not validated here for meningioma, pituitary tumour, metastasis or every intracranial abnormality.
- Regional Grad-CAM covers one crop and does not prove causal or clinically meaningful reasoning.
- The system does not perform registration, DICOM de-identification, authentication, audit logging or regulated deployment.
- Dice alone omits boundary distance, lesion-wise sensitivity, calibration and inter-observer variability.

6.3 Conclusion

SYSTEM.ONCO achieves the project objective: it connects validated multimodal MRI ingestion, a pretrained 3-D CNN, aligned segmentation and explanation outputs, quantitative measurements and interactive WebGL review. The application makes both strong and weak cases visible and preserves the distinction between model probability, reference overlap and clinical interpretation.

6.4 Future Scope

Future work should evaluate a larger patient-level held-out cohort; report Dice, 95% Hausdorff distance, sensitivity, specificity, calibration and lesion-wise errors; compare SegResNet with nnU-Net and transformer baselines under one protocol; add full-volume attribution or uncertainty estimation; and conduct usability testing with qualified reviewers. A separate 2-D classifier may be trained on the documented Kaggle and Mendeley sources, but its claims must remain separate from volumetric segmentation. Any clinical extension requires governance, security, DICOM workflows, prospective validation and medical-device review.

Final interpretation remains the responsibility of qualified medical professionals. This prototype is an educational and research aid, not a diagnostic device.

References

- [1] D. N. Louis et al., “2021 WHO Classification of CNS Tumours: a summary,” *Neuro-Oncology*, vol. 23, pp. 1231–1251, 2021. doi: 10.1093/neuonc/noab106.
- [2] B. H. Menze et al., “The Multimodal Brain Tumor Image Segmentation Benchmark,” *IEEE TMI*, vol. 34, pp. 1993–2024, 2015. doi: 10.1109/TMI.2014.2377694.
- [3] S. Bakas et al., “TCGA glioma MRI collections with expert labels,” *Scientific Data*, vol. 4, 170117, 2017. doi: 10.1038/sdata.2017.117.
- [4] U. Baid et al., “BraTS 2021 Benchmark,” arXiv:2107.02314, 2021.
- [5] O. Ronneberger et al., “U-Net,” *MICCAI*, 2015. doi: 10.1007/978-3-319-24574-4_28.
- [6] Ö. Çiçek et al., “3D U-Net,” *MICCAI*, 2016. doi: 10.1007/978-3-319-46723-8_49.
- [7] F. Milletari et al., “V-Net,” *3DV*, 2016. doi: 10.1109/3DV.2016.79.
- [8] S. Pereira et al., “Brain Tumor Segmentation Using CNNs in MRI,” *IEEE TMI*, vol. 35, pp. 1240–1251, 2016. doi: 10.1109/TMI.2016.2538465.
- [9] K. Kamnitsas et al., “Efficient multi-scale 3D CNN for brain lesion segmentation,” *Medical Image Analysis*, vol. 36, pp. 61–78, 2017. doi: 10.1016/j.media.2016.10.004.
- [10] A. Myronenko, “3D MRI brain tumor segmentation using autoencoder regularization,” *BrainLes*, 2018. arXiv:1810.11654.
- [11] F. Isensee et al., “nnU-Net,” *Nature Methods*, vol. 18, pp. 203–211, 2021. doi: 10.1038/s41592-020-01008-z.
- [12] R. R. Selvaraju et al., “Grad-CAM,” *ICCV*, pp. 618–626, 2017.
- [13] M. J. Cardoso et al., “MONAI,” arXiv:2211.02701, 2022.
- [14] A. Paszke et al., “PyTorch,” *NeurIPS* 32, pp. 8024–8035, 2019.
- [15] R. W. Cox et al., “NIfTI-1,” *OHBM*, 2004. Available: nifti.nimh.nih.gov.
- [16] NiiVue Contributors, “NiiVue: a WebGL2 medical image viewer,” *Zenodo*, 2021. doi: 10.5281/zenodo.5786270.
- [17] MedOtter, “BraTS 2023 GLI Dataset,” *Hugging Face Datasets*, 2023.
- [18] MONAI, “BraTS MRI Segmentation: pretrained SegResNet bundle,” *Hugging Face*, 2024.
- [19] M. Nickparvar, “Brain Tumor MRI Dataset,” *Kaggle*, ver. 2, 7,200 images, CC BY 4.0.
- [20] M. I. K. Hira et al., “Brain Tumor MRI Dataset,” *Mendeley Data*, ver. 1, 2025. doi: 10.17632/zwr4ntf94j.1.
- [21] J. Cheng et al., “Brain tumor dataset,” *Figshare*, ver. 5, 2017. doi: 10.6084/m9.figshare.1512427.v5.
- [22] K. H. Zou et al., “Validation metrics for probabilistic brain tumour segmentation,” *Statistics in Medicine*, vol. 23, pp. 1259–1282, 2004. doi: 10.1002/sim.1728.
- [23] O. Kouli et al., “Automated brain tumor identification using MRI: systematic review and meta-analysis,” *Neuro-Oncology Advances*, 2022. doi: 10.1093/noajnl/vdac081.
- [24] T. A. Soomro et al., “Image Segmentation for MR Brain Tumor Detection Using Machine Learning: A Review,” *IEEE Rev. Biomed. Eng.*, vol. 16, pp. 70–90, 2023. doi: 10.1109/RBME.2022.3185292.
- [25] S. Bouhafra and H. El Bahi, “Deep Learning Approaches for Brain Tumor Detection and Classification Using MRI,” *J. Imaging Inform. Med.*, vol. 38, pp. 1403–1433, 2025. doi: 10.1007/s10278-024-01283-8.